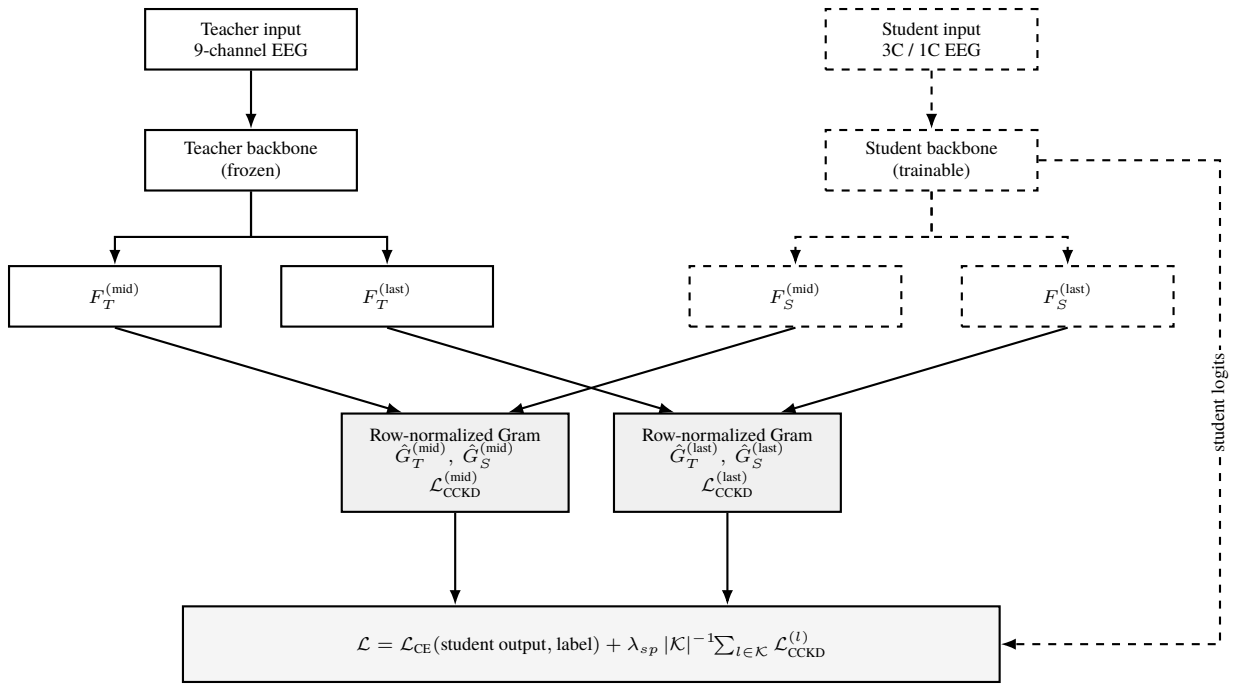
Teacher input
9-channel EEG
Teacher backbone
(frozen)
$F_T^{(\text{mid})}$
$F_T^{(\text{last})}$
Student input
3C / 1C EEG
Student backbone
(trainable)
$F_S^{(\text{mid})}$
$F_S^{(\text{last})}$
Row-normalized Gram
$\hat{G}_T^{(\text{mid})}$, $\hat{G}_S^{(\text{mid})}$
$\mathcal{L}_{\text{CCKD}}^{(\text{mid})}$
Row-normalized Gram
$\hat{G}_T^{(\text{last})}$, $\hat{G}_S^{(\text{last})}$
$\mathcal{L}_{\text{CCKD}}^{(\text{last})}$
student logits
$\mathcal{L} = \mathcal{L}_{\text{CE}}(\text{student output, label}) + \lambda_{sp}\,|\mathcal{K}|^{-1}\sum_{l\in\mathcal{K}}\mathcal{L}_{\text{CCKD}}^{(l)}$